

Saurav Koduri

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EDUCATION

University of Illinois Urbana-Champaign

Major: BS, Electrical Engineering; Cumulative GPA: 3.91/4.00

Class of 2028

Courses: Digital Systems, Analog Signal Processing, Computer Systems and Programming, Linear Algebra, Differential Equations, Discrete Structures, Intro to Computing/Electronics, Physics Mech/EM, Quantum Physics, Calculus III

EXPERIENCE

Humanoid Electrical Team Project Lead

Fall 2025 – Present

UIUC SigRobotics

- Designing a custom four-layer controller and power-distribution PCB in KiCad for a five-finger robotic hand using an ESP32-S3 and 10 DYNAMIXEL XL330-M288-T actuators, supporting coordinated real-time joint control.
- Implementing a shared 3.3V TTL half-duplex communication interface, native USB-C programming, boot/reset circuitry, ESD protection, and expansion interfaces for reliable firmware development and sensor integration.
- Engineering a protected 3S battery power architecture with separate logic and motor rails, including a high-current 5V converter targeting 16A continuous output, power budgeting, component selection, and copper planning.

Sage Program Researcher

Summer 2026

Argonne National Lab

- Built an edge-AI wildlife detection pipeline that ingests image data from distributed camera sensors and combines YOLO11 for animal localization, BioCLIP for species identification, and Gemma 3 for safe/dangerous classification.
- Developed multi-frame animal tracking with persistent track IDs, confidence filtering, and track-expiration logic to identify newly arriving animals, maintain identities across camera frames, and prevent duplicate alerts.
- Integrated Twilio SMS and Slack notifications with annotated images, danger scores, and timestamps.
- Containerized the application for NSF SAGE edge-computing infrastructure using Docker, PyWaggle, and SAGE job manifests; integrated camera-image ingestion, persistent tracker state, and fault-tolerant CSV and annotated-image outputs that continue operating when external alert services are unavailable.

Machine Learning Intern

Summer 2025

Sizzle.gg

- Built Python scripts to curate 100,000+ gameplay images from livestreams for ML model training and evaluation.
- Built and refined an image classification model to identify the game being played on screen, detecting 7 games and expanding toward 50+ games for broader livestream coverage and automated tagging.
- Migrated video game streaming data from MongoDB to PostgreSQL by designing a relational schema and mapping data structures to preserve integrity and support reliable downstream queries and analysis.

Project Associate

Spring 2026 – Present

Illinois Enactus

- Building a Q'anjob'al language-access platform in Champaign to reduce communication barriers.
- Developed multilingual accessibility features using user-centered design, language-support workflows, and technical problem-solving to improve platform usability, community communication, and information access.

PROJECTS

Autonomous Agricultural Tractor System

Summer 2026

- Developing AI perception models on a ROS-based tractor to autonomously navigate a pecan farm, upgrading from YOLO to AlpaMayo models to improve perception accuracy and autonomous driving performance.
- Migrating the tractor's 192-node ROS 1 system and supporting packages to ROS 2, updating interfaces and dependencies for a more modular AI framework and improved scalability.
- Building an NVIDIA Isaac Sim tractor simulation to test autonomy software and sensor configurations.

Resistive-Key Vehicle Safety System

Spring 2026

- Created an analog authentication circuit that converts a resistor-based key value into a target voltage
- Integrated comparator circuits and AND-gate logic to validate the key voltage, then implemented a three-second delay circuit to slow repeated resistance attempts to reduce successful hacker attempts

TECHNICAL SKILLS

Software/Hardware: C/C++, Python, Linux, SQL, SolidWorks, Matlab, ROS 2, Isaac Sim, KiCad, PyTorch

Interests: Basketball (Golden State Warriors), Tennis, Saxophone, Skiing, Rush Hour, Deep-Dish Pizza